

Course Title : Database Management Systems	
Course Code :	No of Credits :
Faculty Name :	Course Teacher name :
School Name :	Email id :
	Contact number :
Pre-requisites :	
Learning Weeks :	
Student Intake :	
Course Type :	
Course Difficulty Level :	
Offered in Academic Year :	Semester : 4
Offered in Program :	

COURSE DESCRIPTION

This course covers the principles and implementation of database systems. It includes database architecture, ER modeling, relational model, SQL, normalization, transactions, concurrency control, indexing, query optimization, and storage mechanisms. The course emphasizes both conceptual foundations and hands-on implementation using MySQL.

COURSE OUTCOMES

After completion of this course, the learners will be able to:

SI No	Description	Bloom's Level
CO1	Explain DBMS architecture, ER modeling, relational model concepts, and ER-to-relational mapping.	2
CO2	Construct and manipulate databases using SQL including DDL, DML, joins, and subqueries.	3
CO3	Develop advanced SQL solutions using views, CTEs, stored procedures, and triggers.	4
CO4	Analyze functional dependencies, perform normalization, and evaluate transaction and	4

	concurrency mechanisms.	
CO5	Evaluate indexing strategies, query optimization techniques, and storage mechanisms for performance tuning.	5

** 1) Remember, 2) Understand, 3) Apply, 4) Analyze, 5) Evaluate, 6) Create

MAPPING of COs and POs

COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2														
CO3														
CO4														
CO5														

Note: Degree of relevance/correlation: 1-Low, 2-Medium, 3-High, Blank or '-' - Nil

SYLLABUS

Content	Contact Hours	CO Mapping	Week
Unit I Database systems vs file systems, data redundancy and anomalies, DBMS functions, three-schema architecture, database users and DBMS internals, ER modeling concepts, entities and attributes, keys and relationships, cardinality and participation constraints, weak entities, specialization and aggregation, relational model concepts, integrity constraints, relational algebra operations, ER-to-relational mapping for strong and weak entities, mapping relationships and specialization strategies		CO1	
Unit II SQL fundamentals, DDL and DML commands, constraints and data types, SELECT queries with filtering and sorting, aggregate		CO2	

functions and GROUP BY, HAVING clause and CASE expressions, string and date functions, joins including inner and outer joins, self joins and multi-table joins,			
Unit III Subqueries and correlated subqueries, EXISTS and NOT EXISTS, set operations, Views and Common Table Expressions, recursive CTEs, stored procedures and stored functions, parameters and control flow constructs, error handling mechanisms, triggers with BEFORE and AFTER operations, OLD and NEW references, constraint enforcement and validation logic		CO3	
Unit IV Functional dependencies and attribute closure, candidate keys and Armstrong's axioms, normalization concepts from 1NF to BCNF, lossless join and dependency preservation, denormalization tradeoffs, transaction management and ACID properties, isolation levels and concurrency issues, locking protocols and deadlock handling, recovery mechanisms and logging concepts		CO4	
Unit V Indexing fundamentals and B+ trees, clustered and secondary indexes, composite and covering indexes, query optimization and EXPLAIN analysis, cost-based optimization and query rewriting, file organization techniques, storage engines and InnoDB architecture, buffer management and disk I/O concepts		CO5	

PRACTICALS

EXPERIMENT	DESCRIPTION	CO MAPPING
Experiment 1	Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.	CO1
Experiment 2	Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.	CO1

Experiment 3	Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.	CO2
Experiment 4	Using the Employee schema, write queries with INNER JOIN, LEFT JOIN, self-join, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare execution plans using EXPLAIN.	CO2
Experiment 5	Create SQL views for department salary summary and employee hierarchy. Test updatability of views. Implement a recursive CTE to display reporting chains.	CO3
Experiment 6	Create a stored procedure <code>transfer_employee(emp_id, new_dept_id)</code> with validation and error handling. Implement triggers for salary validation and audit logging. Test edge cases.	CO3
Experiment 7	Given a denormalized Sales table, identify functional dependencies and normalize step-by-step from UNF to 3NF and BCNF. Verify lossless join and dependency preservation using SQL queries.	CO4
Experiment 8	Implement banking-style transactions using BEGIN, COMMIT, ROLLBACK, and SAVEPOINT. Simulate partial failures. Demonstrate isolation levels and concurrency anomalies using two sessions.	CO4
Experiment 9	Create a Railway Booking table with at least 5,000 records. Benchmark query performance before and after creating single and composite indexes. Use EXPLAIN to compare execution plans.	CO5
Experiment 10	Generate a dataset with at least 10,000 rows. Write five slow queries, analyze using EXPLAIN, optimize using indexing and query rewriting, and compare execution time before and after optimization.	CO5

TEXTBOOKS/REFERENCE BOOKS

- Elmasri, R., & Navathe, S. B. — Fundamentals of Database Systems, Pearson.
- Ramakrishnan, R., & Gehrke, J. — Database Management Systems, McGraw-Hill.
- Date, C. J. — An Introduction to Database Systems, Pearson.
- Garcia-Molina, H., Ullman, J. D., & Widom, J. — Database Systems: The Complete Book, Pearson.